Experiment No. 5

Fourier Series Analysis of Periodic Signals

5.1.To analyze Fourier series of continuous-time periodic signals and analyze their waveform Characteristics.

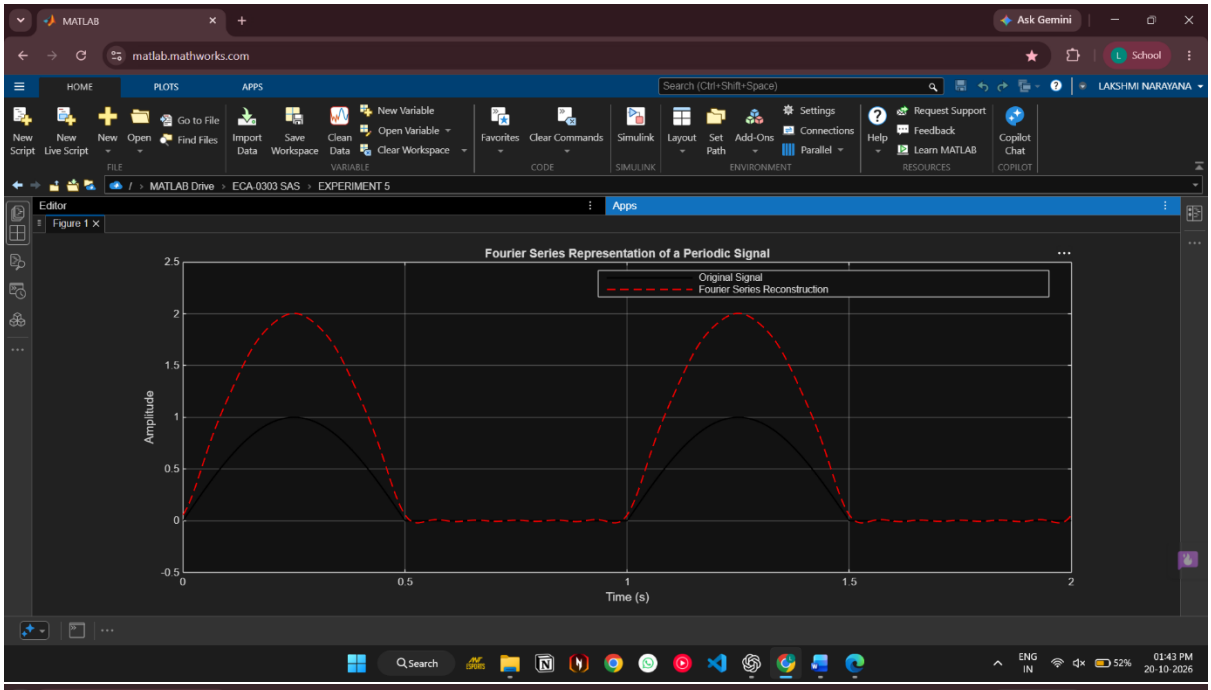
Experimental Procedure

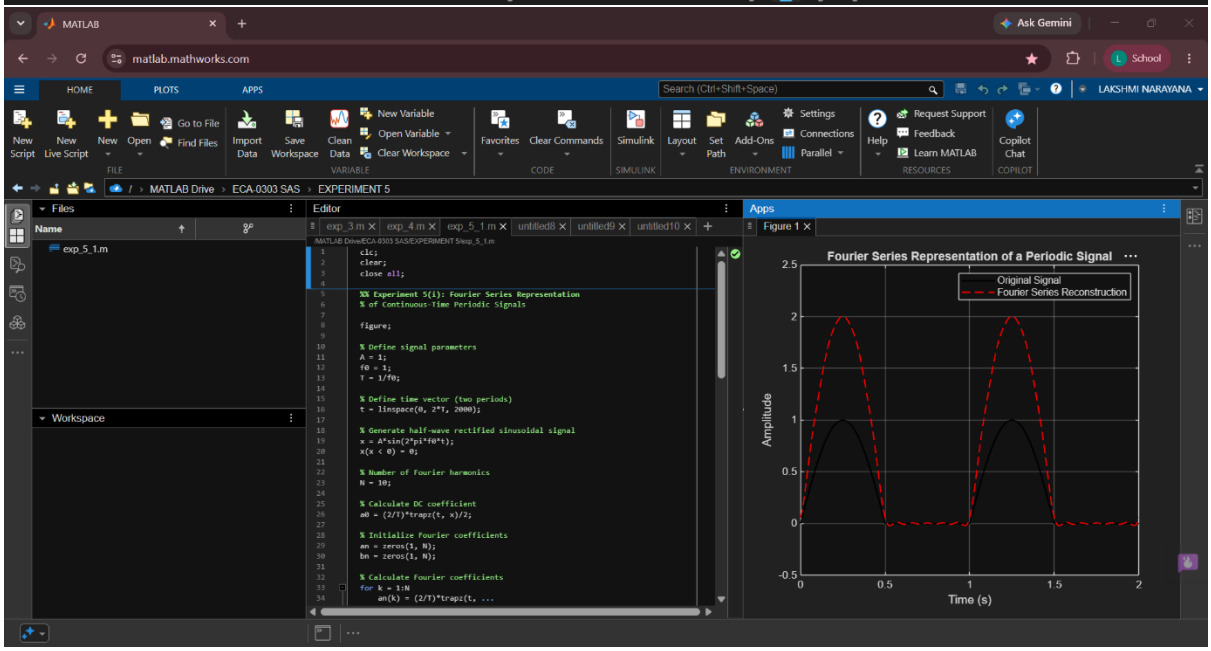
1. Define the time vector and fundamental frequency.
2. Generate the half-wave rectified sinusoidal periodic signal.
3. Compute the Fourier coefficients up to N = 10.
4. Construct the Fourier-series approximation.
5. Plot the original and reconstructed signals and compare them.

MATLAB Coding

```
clc; clear; close all;
%% 5(i) Analysis of Fourier Series Representation of Continuous-Time Periodic Signals
figure;
% Fourier Series Representation of a Periodic Signal
A = 1;
f0 = 1;
T = 1/f0;
t = linspace(0, 2*T, 2000);
% Half-wave rectified sinusoidal signal
x = A*sin(2*pi*f0*t);
x(x < 0) = 0;
N = 10;
a0 = (2/T)*trapz(t, x)/2;
an = zeros(1,N);
bn = zeros(1,N);
for k = 1:N
    an(k) = (2/T)*trapz(t, x.*cos(2*pi*k*f0*t));
    bn(k) = (2/T)*trapz(t, x.*sin(2*pi*k*f0*t));
end
```

Output:


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New Script
New Live Script
New
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Go to File
Find Files
Import Data
Save Workspace
Clean Data
New Variable
Open Variable
Clear Workspace
Favorites
Clear Commands
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Set Path
Add-Ons
Settings
Connections
Parallel
Help
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Editor
Apps
Figure 1
Fourier Series Representation of a Periodic Signal
Original Signal
Fourier Series Reconstruction
Amplitude
Time (s)
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Files
Name
exp_5_1.m
Workspace
Editor
exp_3.m
exp_4.m
exp_5_1.m
untitled8
untitled9
untitled10
clc;
clear;
close all;
%% Experiment 5(i): Fourier Series Representation
% of Continuous-Time Periodic Signals
figure;
% Define signal parameters
A = 1;
f0 = 1;
T = 1/f0;
% Define time vector (two periods)
t = linspace(0, 2*T, 2000);
% Generate half-wave rectified sinusoidal signal
x = A*sin(2*pi*f0*t);
x(x < 0) = 0;
% Number of Fourier harmonics
N = 10;
% Calculate DC coefficient
a0 = (2/T)*trapz(t, x)/2;
% Initialize Fourier coefficients
an = zeros(1, N);
bn = zeros(1, N);
% Calculate Fourier coefficients
for k = 1:N
an(k) = (2/T)*trapz(t, ...
Apps
Figure 1
Fourier Series Representation of a Periodic Signal
Original Signal
Fourier Series Reconstruction
Amplitude
Time (s)